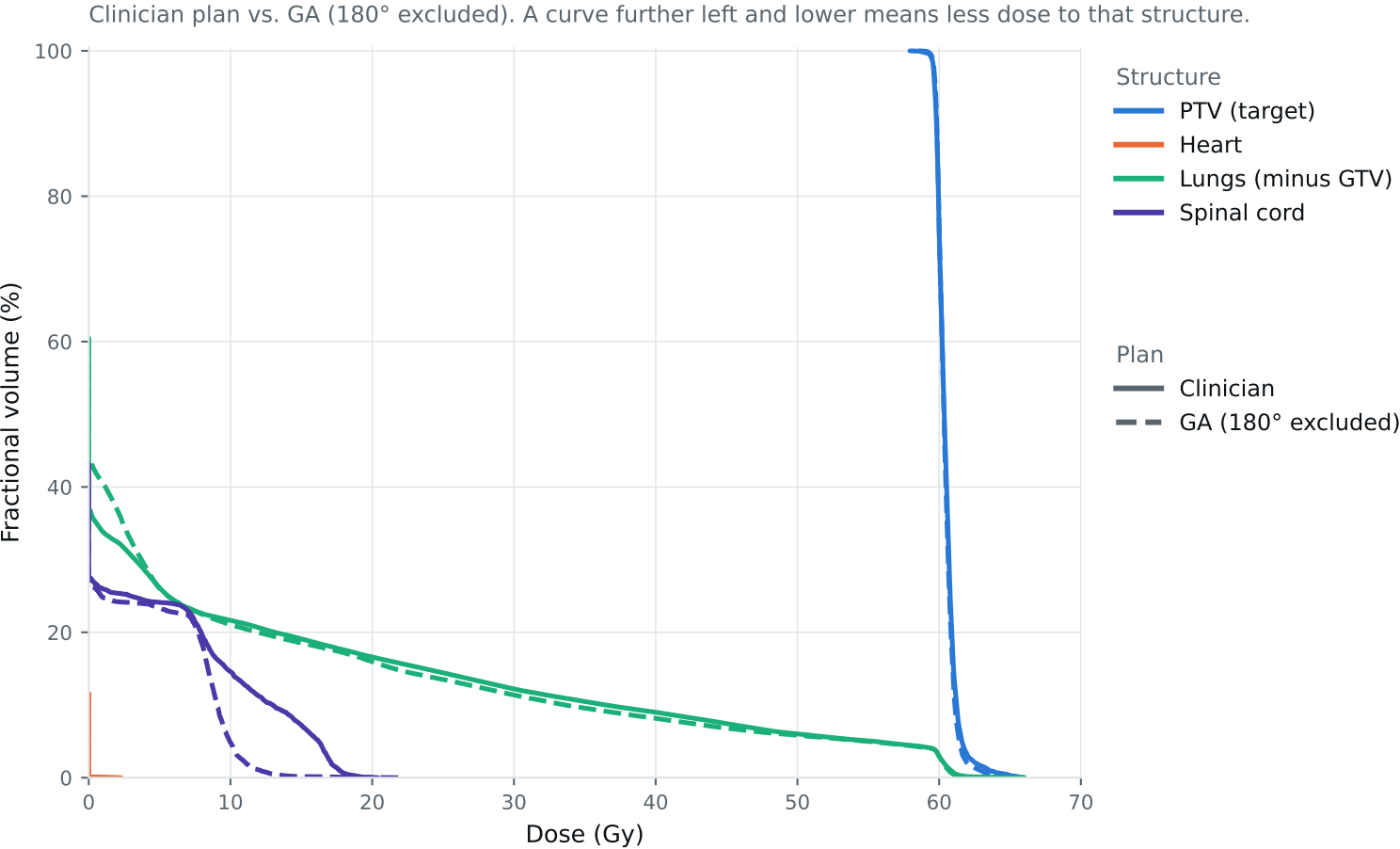


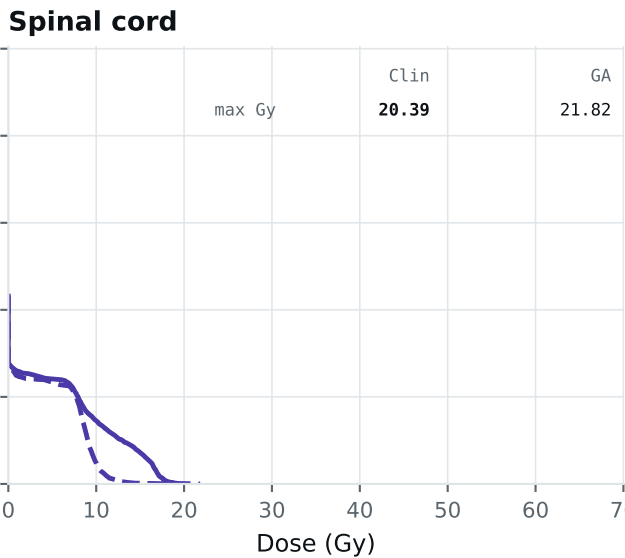
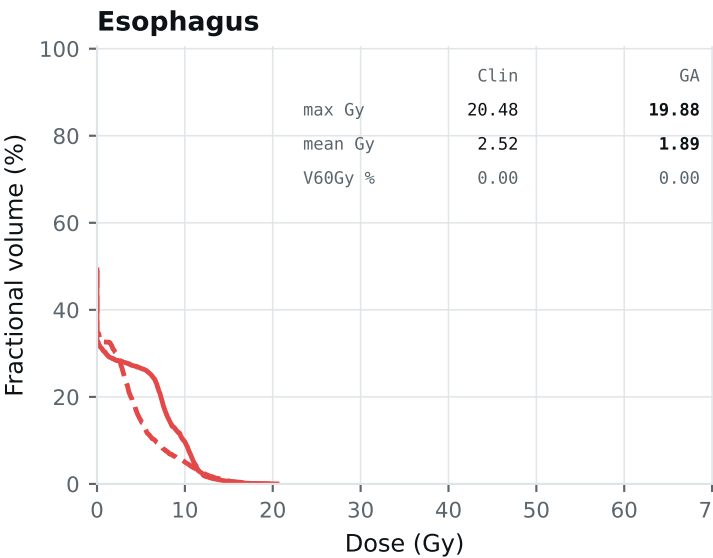
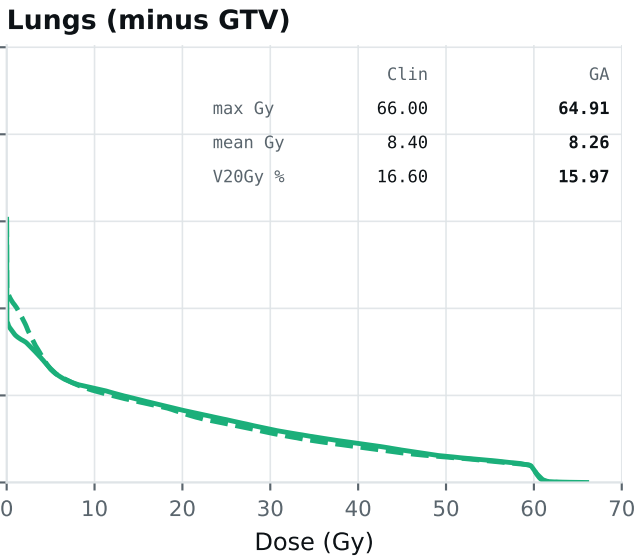
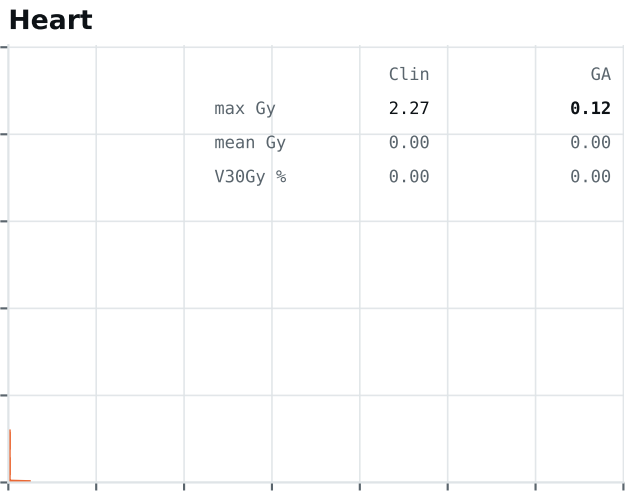
# Dose-volume histogram — Lung\_Patient\_16



Curves further left are better for organs; the PTV curve should stay high, then drop sharply. Target curves should overlap; organ curves show the tradeoffs. Esophagus is on page 2.

# DVH by structure — Lung\_Patient\_16

Same curves, one structure per panel, all plans, each annotated with that structure's protocol metrics.  
Bold = clinically preferable: organs favor lower dose; PTV favors the value closest to goal within the limit. Grey = all plans are within 0.01 of each other. Amber misses a goal,



Plan  
— Clinician  
- - GA (180° excluded)

# Clinical criteria — Lung\_Patient\_16

Organs: lower is better. PTV: closest to goal without exceeding the limit. Bold marks the best plan.

| Criterion                       | Limit     | Goal                                    | Clinician    | GA<br>180° excluded |    |
|---------------------------------|-----------|-----------------------------------------|--------------|---------------------|----|
| GTV max                         | 69        | 66                                      | 62.11        | <b>62.05</b>        | Gy |
| PTV max                         | 69        | 66                                      | <b>66.00</b> | 64.91               | Gy |
| ESOPHAGUS max                   | 66        | —                                       | 20.48        | <b>19.88</b>        | Gy |
| ESOPHAGUS mean                  | 34        | 21                                      | 2.52         | <b>1.89</b>         | Gy |
| ESOPHAGUS V60Gy                 | 17        | —                                       | 0.00         | 0.00                | %  |
| HEART max                       | 66        | —                                       | 2.27         | <b>0.12</b>         | Gy |
| HEART mean                      | 27        | 20                                      | 0.00         | 0.00                | Gy |
| HEART V30Gy                     | 50        | 48                                      | 0.00         | 0.00                | %  |
| LUNG_L max                      | 66        | —                                       | <b>9.53</b>  | 34.55               | Gy |
| LUNG_R max                      | 66        | —                                       | 66.00        | <b>64.91</b>        | Gy |
| CORD max                        | 50        | 48                                      | <b>20.39</b> | 21.82               | Gy |
| SKIN max                        | 60        | —                                       | <b>38.93</b> | 40.82               | Gy |
| LUNGS_NOT_GTV max               | 66        | —                                       | 66.00        | <b>64.91</b>        | Gy |
| LUNGS_NOT_GTV mean              | 21        | 20                                      | 8.40         | <b>8.26</b>         | Gy |
| LUNGS_NOT_GTV V20Gy             | 37        | —                                       | 16.60        | <b>15.97</b>        | %  |
| PTV D95 (target coverage)       |           |                                         | 59.70        | 59.73               |    |
| Objective value (search signal) |           |                                         | 59.58        | 53.13               |    |
| Pipeline time (setup + solve)   |           |                                         | 17 s         | 18 s                |    |
| Beam angles (gantry°)           | Clinician | 0°, 334°, 310°, 245°, 212°, 185°        |              |                     |    |
|                                 | GA        | 185°, 30°, 225°, 255°, 270°, 300°, 345° |              |                     |    |

Grey rows: all plans agree within 0.01 — heart, lung and LUNGS\_NOT\_GTV max sit at 66 Gy because those structures abut the target. Amber misses the goal; red exceeds the limit.

Source: Lung\_Patient\_16\_clinical\_metrics\_full\_resolution.json, full resolution. The curves and table come from the same solve.